3. OSI vs TCP/IP Model

Both models explain how data travels from one device to another. OSI is the theoretical reference model with 7 layers. TCP/IP is the practical model the internet actually runs on, with 4 layers.

OSI layers with protocols and an attack for each

Layer	Name	What it does	Example protocols	One attack
7	Application	Interface for user apps	HTTP, HTTPS, FTP, SMTP, DNS	SQL injection / phishing
6	Presentation	Encryption, encoding, compression	SSL/TLS, JPEG, ASCII	SSL stripping
5	Session	Opens, manages and closes sessions	NetBIOS, RPC, PPTP	Session hijacking
4	Transport	End-to-end delivery, ports, reliability	TCP, UDP	SYN flood
3	Network	Logical addressing and routing	IP, ICMP, IPsec	IP spoofing / ICMP flood
2	Data Link	MAC addressing, frames, switching	Ethernet, ARP, PPP	ARP spoofing / MAC flooding
1	Physical	Raw bits over cable or wireless	Cables, hubs, Wi-Fi signals	Wiretapping / cable cutting

TCP/IP layers and how they map

TCP/IP layer	Matching OSI layers	Protocols
Application	5, 6, 7	HTTP, DNS, FTP, SMTP, SSH
Transport	4	TCP, UDP
Internet	3	IP, ICMP
Network Access	1, 2	Ethernet, Wi-Fi, ARP

OSI vs TCP/IP at a glance

Point	OSI	TCP/IP
Layers	7	4
Type	Theoretical reference	Practical, used on the internet
Developed by	ISO	US Department of Defense
Approach	Protocol independent	Built around specific protocols
Session and presentation	Separate layers	Merged into application

Easy way to remember

- OSI top to bottom: **A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing
- A SOC analyst uses this daily. If an alert says "SYN flood", you know it is a transport layer problem.